

Real Time Stock Market Analysis

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Abstract—The stock market is extremely important in the financial component of the country's growth, but it is also extremely dynamic and unpredictable. Significant political concerns, investor calls, journal stories, the company's restructuring and development strategies, and many other factors will quickly affect it. Single business day, thousands of investors invest in the stock market. Some of these investors either lose or gain money. However, the loss or benefit on any given trading day is unpredictable. The demand for stock price predictions is incredibly strong, necessitating the use of stock market research. Effective forecast tools assist traders indirectly by supplying supportive knowledge such as price position in the future. In this paper, the system for forecasting stock is developed that makes use of stacked LSTM, linear regression, random forest and K-nearest neighbours neural network algorithm, to forecast the stock trends on the basis of the price history. The proposed framework is designed to analyse every company's stock using mathematical technical metrics. The remaining metrics are probabilistic, although others are deterministic. The aim of this paper is to reduce the risk of failure in each exchange while increasing benefit.

Keywords—LSTM, Linear Regression, Random Forest, K-Nearest Neighbours, Neural Network

I. INTRODUCTION

The planet is awash in data, and we are only interested in the data that is important to us. Technology, arts, technology, history, culture, geography, and economics are just some of the areas where knowledge can be categorized. In certain circumstances, it is important for us to distil the knowledge that is applicable to us. The information available, especially in the field of finance, is not well structured, and there is only information about the current situation. The stock market can be thought of as a platform where nearly all of the world's main economic transactions take place at a volatile pace known as the stock valuation, which is dependent on market equilibrium. Most investors want to hear about the market's potential situation in order to make good investments. One popular method is to use ML algorithms to learn from historical market data in order to forecast future prices. This paper focuses on a specific category of Recurrent Neural Network ML algorithms

These algorithms have short-term memory capability and hence perform better than other traditional approaches in the ML field. The algorithms used here are the LSTM (Long-Short term memory), Linear Regression, Random Forest and K-nearest neighbours. The sentimental analysis of the tweets related to the stocks is also done.

The paper is structured as follows: Section II explains the previous work, Section III explains the problem statement, Section IV explains the proposed model, Section V explains the results, and Section VI explains the conclusion.

II. RELATED WORK

The authors in [1] have used pattern matching and BP neural networks to build a stock market forecasting method. The suggested scheme overcomes the local least in the objective position of the Neural Networks predicting objective function's local least and the absence of market shifting odds in the pattern matching system, and takes advantage of these two algorithms' special strengths in stock price forecasting.

The authors in [2] have explained the forecasting of stocks using ML. The majority of stockbrokers use mathematical and fundamental analysis, as well as time series analysis, when making stock forecasts. They have proposed a Machine Learning (ML) technique that will be trained using publicly accessible stock data to obtain information, and then use that knowledge to make an accurate prediction. In this sense, the analysis employs Support Vector Machine (SVM) to forecast equity values using regular and up-to-the-minute frequencies for big and small capitalizations, as well as three different markets.

The authors in [3] have presented a hybridised methodology that incorporates the use of scientific and fundamental analyses of stock market metrics for stock price estimation in the future. The hybridised solution was put to the test using publicly available stock records, and the findings demonstrated a significant change over using only technical research variables. In addition, the hybridised approach's

forecast was found to be satisfactory as a tool for investors to make decisions.

The authors in [4] have proposed a novel way to reduce the probability of capital market investing by forecasting stock returns using a versatile class of machine learning algorithms known as ensemble learning. Model is trained using technical metrics such as the Relative Strength Index (RSI), stochastic oscillator, and others. An ensemble of multiple decision trees was used as the learning paradigm.

The authors in [5] have used the Artificial Neural Network and Random Forest techniques for forecasting the next day's closing price for five firms in various business sectors. The financial data is used to generate new variables, such as the stock's open, large, low, and near values, which are then used as inputs to the model. Normal strategic metrics such as RMSE and MAPE are used to assess the models. These two metrics have low values, indicating that the models are good at forecasting stock closing prices.

The authors in [6] have presented the problem to forecast whether the price of a stock will rise or fall. They've developed an experimental mechanism to determine when stock values would increase or fall in comparison to the price of n days earlier. Random forests and gradient boosted decision trees (using XGBoost) are two algorithms that use ensembles of decision trees to promote this relation.

The authors in [7] have introduced the use of RNN and LSTM in fund management for stock market prediction, taking into account time series historical stock data of stocks in the portfolio.

III. PROBLEM STATEMENT

A classification model is designed based on LSTM networks, Linear Regression, random forest and K-Nearest Neighbours to make assumptions on stock price fluctuations. In other words, it tries to figure out what a stock's price is by comparing the outcomes of the above-mentioned prediction algorithms. The model is regenerated and trained on historical price data every trading day, and it is used to make forecasts.

A. LSTM Model

The long short-term memory (LSTM) architecture is a deep learning architecture that employs a recurrent neural network (RNN). Unlike typical feedforward neural networks, LSTM has feedback relationships. It can handle not only single data points but also whole data sequences [8].

Fig 1 depicts the flow chart of the LSTM model.

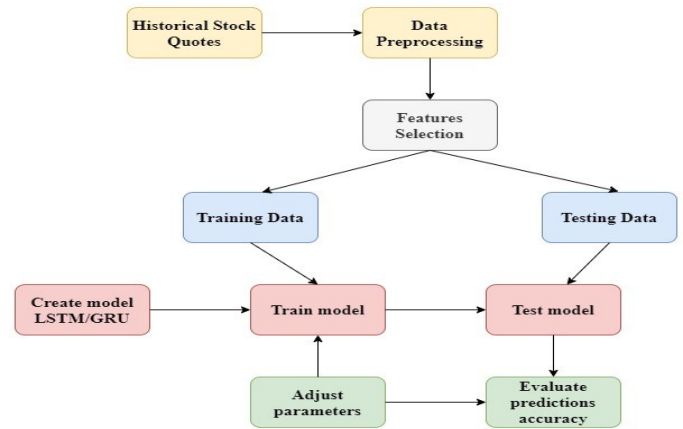


Fig 1 Flow chart of the LSTM model

B. Linear Regression Model

Linear regression can be used to discover a relationship between two or more variables of interest and then make forecasts based on that information. There are only two variables in basic linear regression: one continuous variable and one independent variable. In this algorithm the slope is used to find the predicted stock price [9].

Fig 2 depicts the flowchart of the linear regression model.

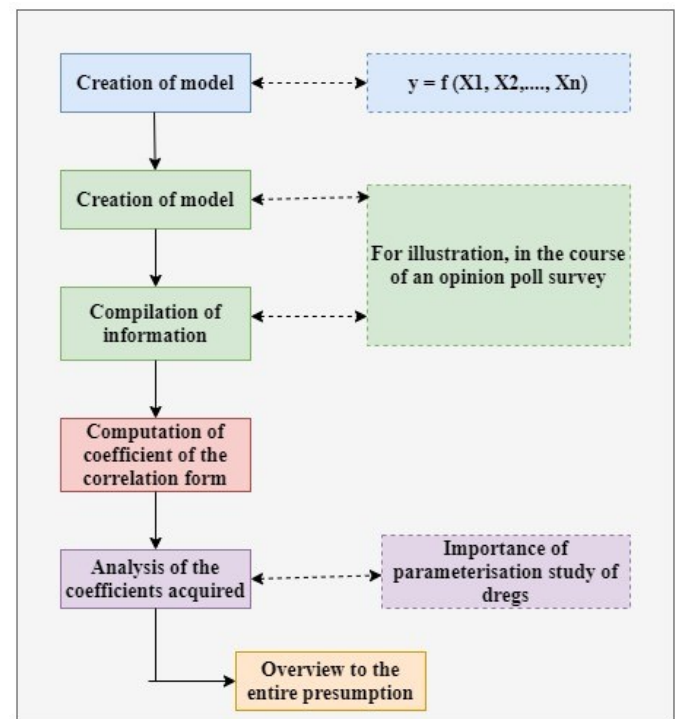


Fig 2 Flowchart of the Linear Regression Model

C. Random Forest Model

The supervised learning technique is used by Random Forest, a well-known machine learning algorithm. In machine learning, it can be used for both classification and regression problems. It is built on ensemble learning, which is a

technique for solving a complex problem by integrating several classifiers and improving the model's accuracy. Rather than relying on a single decision tree, the random forest uses the forecasts from each tree to determine the final performance using the majority of votes [10].

Fig 3 represents the flow chart of the random forest model.

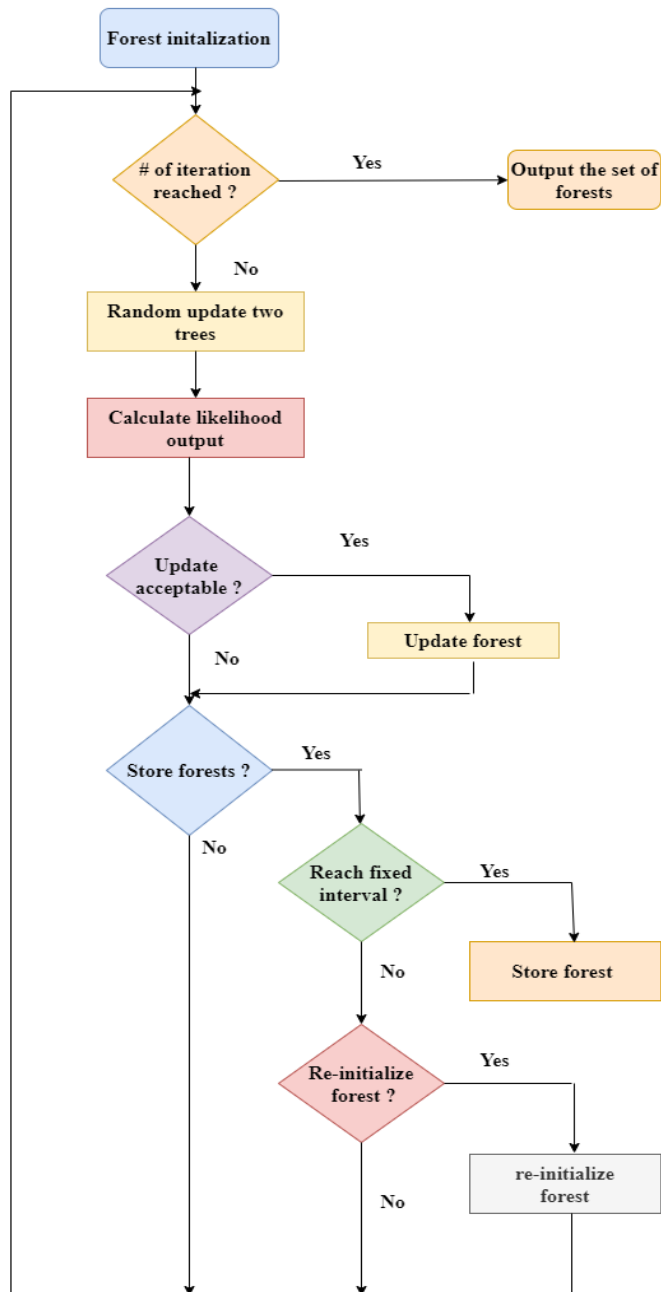


Fig 3 Flowchart of the Random Forest model

D. K Nearest Neighbor Model

The K-Nearest Neighbor algorithm is one of the most simple machine learning algorithms. It is based on the supervised learning approach. It assumes that the new case/data and existing cases are equivalent, so it assigns the new case to a

category that is more similar to the old ones. It keeps all of the existing data and classifies new data points based on their similarity to the existing data [11].

Fig 4 represents the flowchart of the k nearest neighbor model.

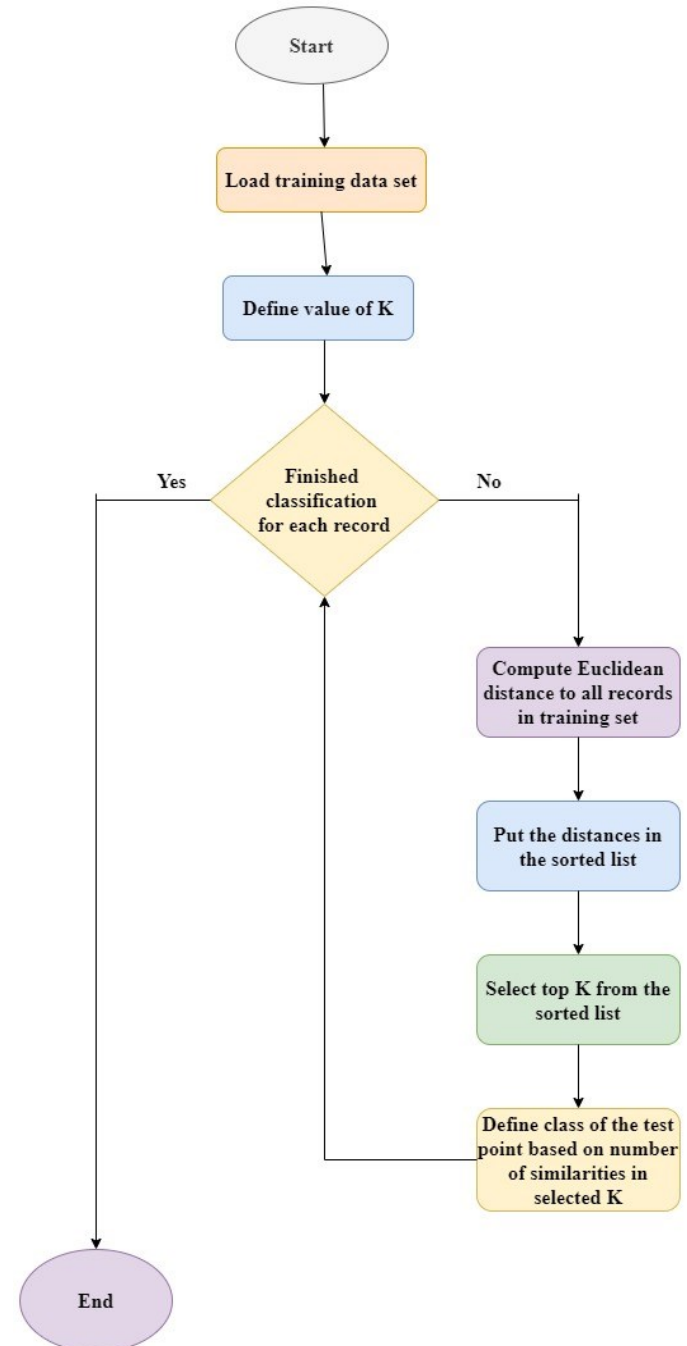


Fig 4 Flowchart of the KNN Model

IV. RESULTS

The apple stock file is used as a dataset [12]. The four different machine learning algorithms that are LSTM, Linear Regression, Random Forest and KNN are used to predict the

stock trend. Fig 5 represents the results from the trained model.



Fig 5 Trained Model Results

In the above figure, the black dots represent the data points and the different colors show the results for different algorithms. All four models are tested simultaneously and the graph is plotted. The test data is tested and performance is analysed with the four machine learning algorithms. Table 1 depicts the mean squared error for the different models.

Table 1 Model Evaluation Table

Model	Mean Squared Error
Linear regression	1.5417086
Random forest	0.316893179
KNN	0.69425064
LSTM	1.689194109

The mean or average of the square of the difference between real and expected values is known as Mean Square Error (MSE) [13]. In the above figure, it can be seen that the random forest model has the least mean squared error which is approximately equal to the 0.31. The lower the mean squared error, the better is the algorithm so it can be concluded that random forest is best among four for the determination of the stock trend.

Fig 6 depicts the prediction of the stock for the next 15 days. The sentimental analysis is also performed on the 1000 tweets about the stock. The sentimental analysis of the tweets tells us about the opinion of the public about the particular stock. After performing the sentimental analysis, it can be seen that 54% tweets are positive and 46% tweets are negative. Fig 7 represents the sentimental analysis for the top 1000 tweets.

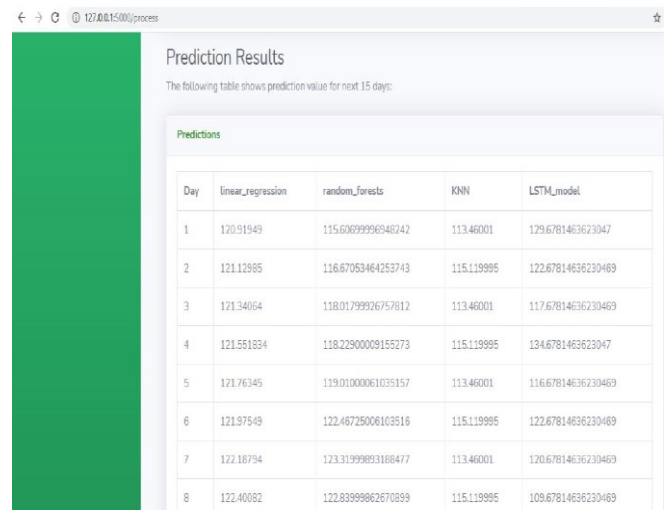


Fig 6 Prediction result of next 15 days

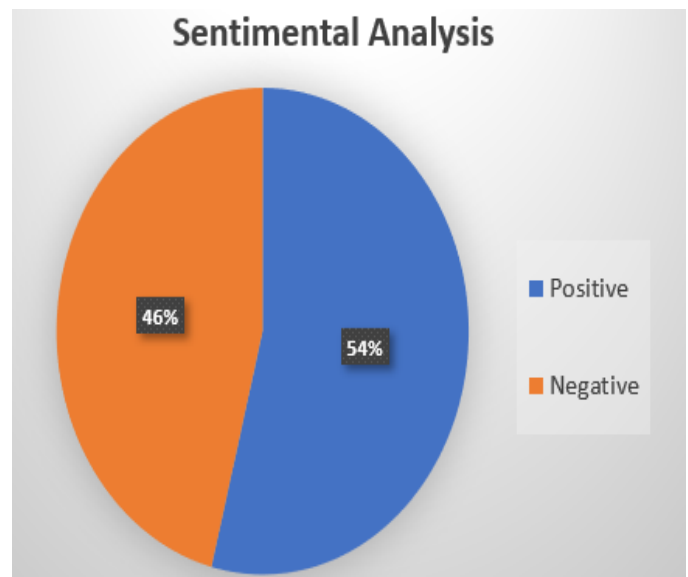


Fig 7 Sentiment analysis pie chart

V. CONCLUSION

Due to its high nonlinear, competitive, and complex existence, predicting the stock market is extremely difficult. Machine learning algorithms, on the other hand, have shown to be successful in stock forecasting in recent years. The growth of stock market investing is increasingly increasing, prompting analysts to develop new approaches for predicting the future using new techniques. The forecasting methodology benefits not only analysts, but also clients and everyone who works in the stock market. A forecasting model with high precision is expected to assist in the prediction of stock indices. In this paper, four models are tested against the stock data set for the prediction. The random forest performs the best as it has the least mean squared error.

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